

Practical Assignment: Full Stack MERN Application

Project Title: Smart Task & Notes Management System

Objective

Build a **full-stack MERN application** where users can:

- Register/Login securely
 - Manage tasks and notes
 - Perform CRUD operations
 - Persist data using MongoDB
 - Deploy the project on cloud platforms
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Module 1: Backend Development (Node.js + Express + MongoDB)

Requirements:

1. **Project Setup**
 - Initialize Node.js project
 - Install dependencies: Express, Mongoose, JWT, bcrypt, cors, dotenv
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2. **Database Design (MongoDB + Mongoose)**

Create schemas:

- **User Schema**
 - name
 - email
 - password (hashed)
 - **Task Schema**
 - title
 - description
 - status (pending/completed)
 - userId (reference)
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3. REST API Development

Implement APIs:

Authentication APIs

- POST /api/auth/register
- POST /api/auth/login

(Use JWT for token generation)

Task APIs (Protected Routes)

- GET /api/tasks → Get all user tasks
 - POST /api/tasks → Create task
 - PUT /api/tasks/:id → Update task
 - DELETE /api/tasks/:id → Delete task
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4. Middleware

- JWT Authentication Middleware
 - Error Handling Middleware
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5. Project Structure (Modular)

```
/project
├ /models
├ /routes
├ /controllers
├ /middleware
└ server.js
```

6. Testing

- Use **Postman**
 - Test all APIs with token authentication
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Module 2: Frontend Development (React)

Requirements:

1. **React Setup**
 - Create React app
 - Install Axios, React Router
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2. **Pages to Develop**
 - Register Page
 - Login Page
 - Dashboard Page
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3. **Features**

Authentication

- Store JWT in Local Storage
 - Protect routes (Private Routes)
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Task Management UI

- Add Task
 - View Tasks
 - Update Task
 - Delete Task
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4. **React Concepts to Use**
 - useState, useEffect
 - Context API (for global auth state)
 - Controlled forms
 - Conditional rendering
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5. **API Integration**
 - Use Axios / Fetch
 - Connect backend APIs
 - Handle loading & error states
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Integration Requirements

- Enable **CORS** in backend
 - Connect frontend with backend API
 - Use `.env` for API URLs
 - Handle token in headers
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Deployment

Deploy:

- **Frontend:** Netlify / Vercel
 - **Backend:** Render / Railway
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Deployment Checklist:

- Environment variables configured
 - MongoDB Atlas connection
 - Backend live API connected to frontend
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Bonus Features (for Extra Marks)

- Search & Filter tasks
 - Dark Mode
 - Due date for tasks
 - Responsive UI
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Submission Requirements

- GitHub Repository (Frontend + Backend)
 - Live deployed links
 - Screenshots of application
 - API testing screenshots (Postman)
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